Volatility Modeling: GARCH vs. Market-Implied Volatility

A Black-Scholes Options Pricing Analysis Across Three Decades

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Dataset & Methodology

This project analyzes 30+ years of daily S&P 500 data to quantify the cost of Black-Scholes' constant-volatility assumption. The primary dataset is SPY (S&P 500 ETF) from Yahoo Finance, covering **7,838 trading days (1993–2024)**, supplemented by QQQ and GLD for cross-asset comparison. Macroeconomic inputs — the CBOE VIX, VIX3M, and 3-Month T-Bill rate — were sourced from the Federal Reserve Economic Data (FRED) API.

The analysis proceeds in four stages: (1) exploratory analysis of return distributions, volatility clustering, and the variance risk premium; (2) fitting GARCH(1,1) and GJR-GARCH(1,1) models to SPY log-returns and validating against model diagnostics; (3) pricing a 30-day at-the-money SPY call option daily under three volatility assumptions — constant historical vol, GJR-GARCH conditional vol, and VIX implied vol — and measuring how prices diverge across four volatility regimes; and (4) benchmarking all three models against the Heston (1993) stochastic volatility framework and testing forecast accuracy against future realized volatility.

Key Findings

Excess Kurtosis (SPY returns)	Days VIX > Realized Vol
10.89	**85.3%**
Avg Variance Risk Premium	Extreme-Regime Mispricing (peak)
3.67 vol pts	**+13.1% of spot**

1. The Normal Distribution Assumption Is Empirically Rejected

SPY log-returns exhibit excess kurtosis of 10.89 and skewness of −0.294 — far outside the normal distribution Black-Scholes assumes. The Jarque-Bera test rejects normality at every conventional significance level ($p \approx 0$). This means crashes occur far more frequently than the model predicts, and any pricing framework built on the normal assumption structurally underestimates tail risk.

2. GJR-GARCH Captures an Asymmetry Standard GARCH Misses

The Glosten-Jagannathan-Runkle extension of GARCH adds a single asymmetric term (γ) that gives extra weight to negative return shocks — the leverage effect. Estimated on SPY, $\gamma = 0.173$ versus $\alpha = 0.007$: a negative shock raises conditional variance roughly **27× more** than an

equivalent positive shock. The model's AIC of 21,128 represents a **257-point improvement** over standard GARCH (21,385), confirming the asymmetry is statistically meaningful, not incidental.

3. VIX Systematically Overestimates Future Volatility

VIX exceeded 21-day realized volatility on 85.3% of trading days, with a persistent average premium of 3.67 annualized volatility points. This gap — the variance risk premium — is not a forecasting error. It reflects structural compensation demanded by options sellers for bearing convex volatility exposure. Interestingly, despite this upward bias, VIX still outperforms constant-vol forecasts on RMSE (7.90% vs. 10.39%), though GJR-GARCH is the most accurate predictor overall (RMSE = 7.67%, bias = +0.35%).

4. Black-Scholes Mispricing Is Regime-Dependent and Material

Using a fixed historical volatility input, Black-Scholes *underprices* options in calm markets (Low regime: −1.30% of spot on average) and *overprices* them during stress (Extreme regime: +3.67% average, +13.1% peak). This systematic, regime-dependent bias grows precisely when accurate pricing matters most — during periods of acute market stress when options are most actively used for protection.

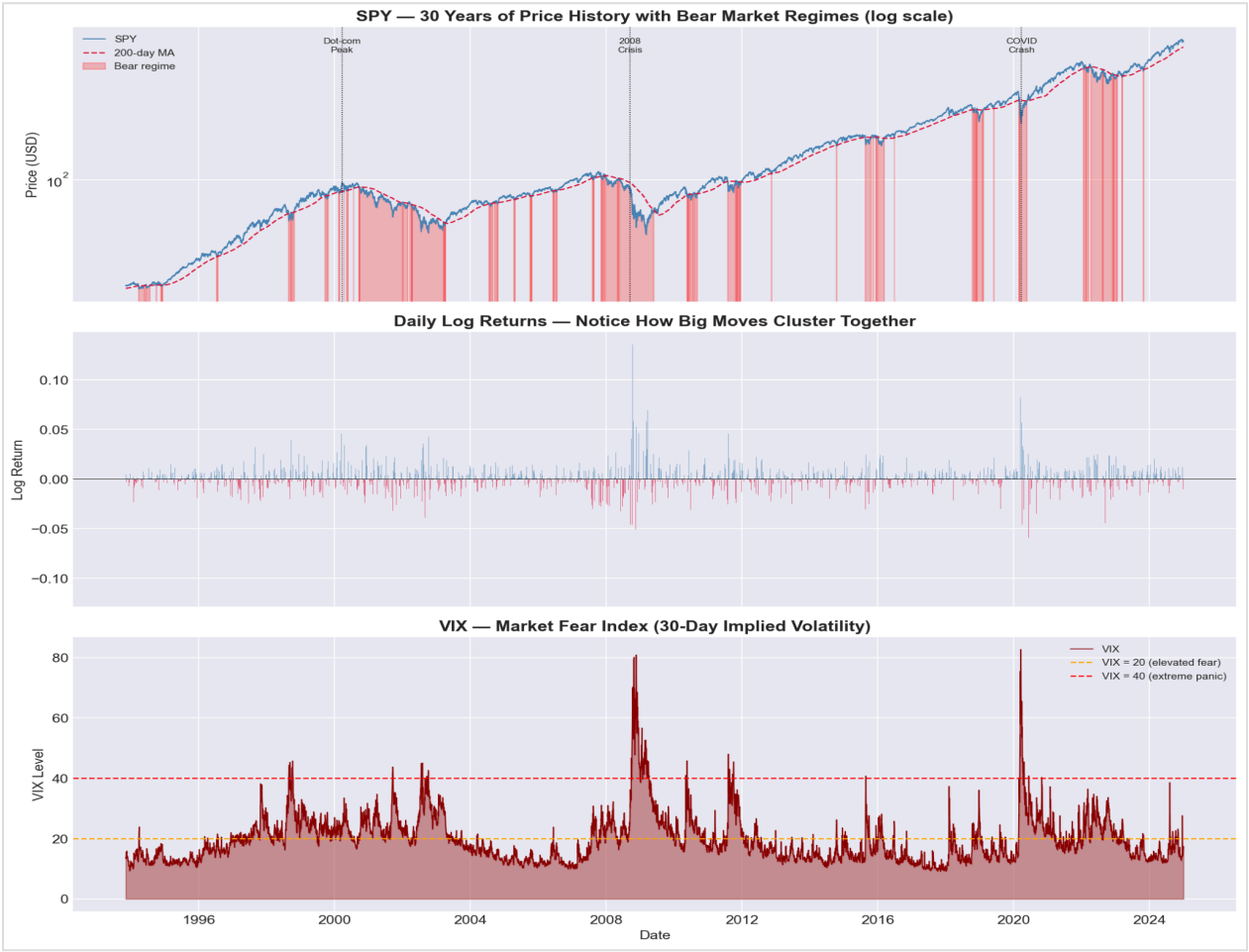

Figure 1. *SPY price history (1993–2024) with bear market regimes shaded (periods below the 200-day moving*

average). SPY spent 23.2% of all trading days in a bear regime, with a maximum drawdown of –55.2% during the 2008 financial crisis.

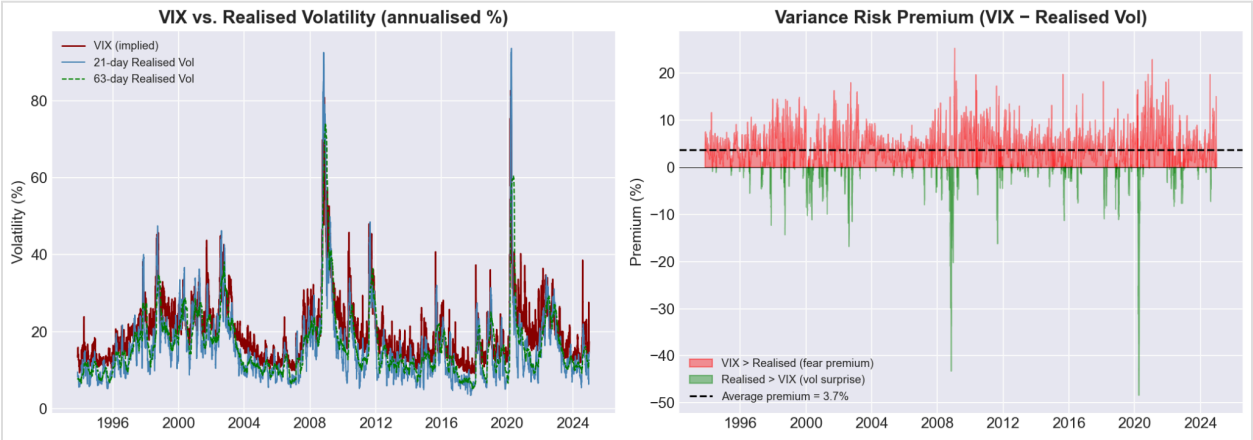

Figure 2. *VIX (implied volatility) vs. 21-day realized volatility, with the variance risk premium shaded. VIX exceeded realized vol on 85.3% of trading days, with an average premium of 3.67 annualized volatility points — a persistent and structurally rational feature of equity options markets.*

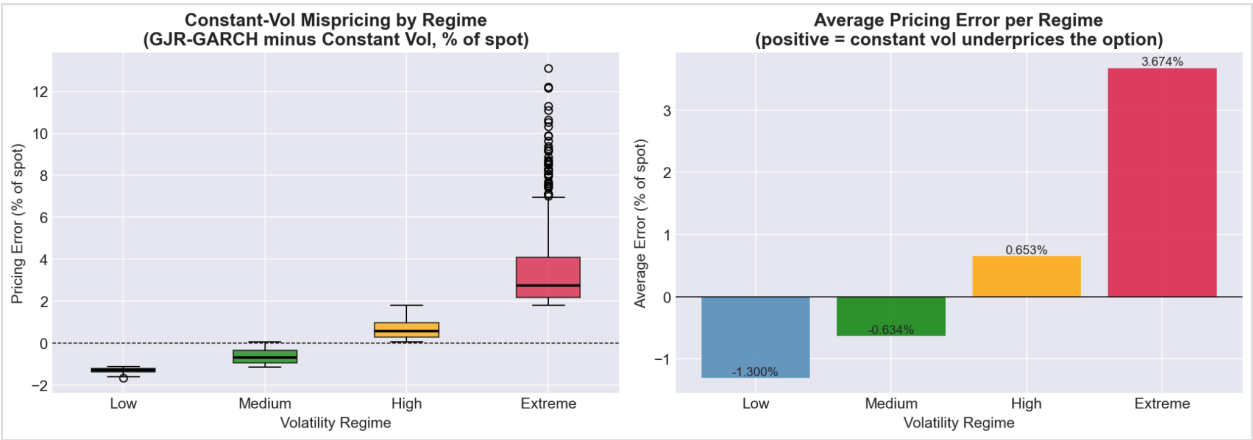

Figure 3. *Black-Scholes mispricing (constant vol vs. GJR-GARCH) across four volatility regimes. The constant-vol model underprices during calm periods and overprices during stress, with average error increasing from –1.30% in the Low regime to +3.67% in the Extreme regime, peaking at +13.1% during crisis periods.*

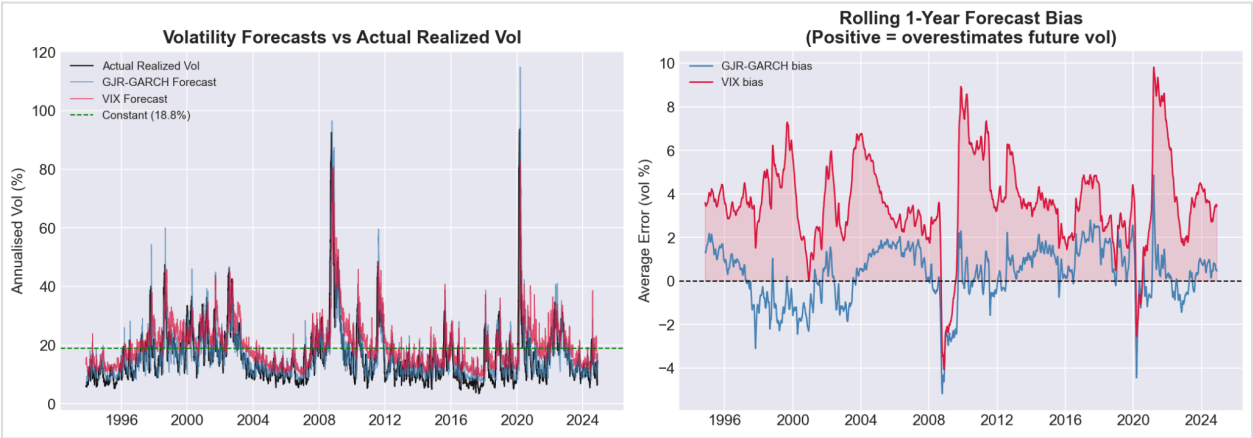

Figure 4. *Volatility forecast accuracy: each model's daily estimate vs. realized volatility over the following 21 trading days. GJR-GARCH achieves the lowest RMSE (7.67%) and near-zero bias (+0.35%), outperforming both VIX (RMSE = 7.90%, bias = +3.66%) and constant vol (RMSE = 10.39%, bias = +2.74%).*

Reflection

The most unexpected finding was how well GJR-GARCH forecasts future realized volatility relative to VIX. Intuitively, VIX — as the market's consensus implied volatility — should aggregate the most information. Instead, its persistent upward bias (averaging +3.66 vol points) makes it a less accurate predictor of what volatility actually turns out to be. GJR-GARCH, updating only from past return data, achieves essentially unbiased forecasts. This reinforces that the variance risk premium is a genuine risk premium, not a forecasting mistake: VIX is not trying to predict realized vol, it is pricing the cost of volatility insurance.

A natural extension would be calibrating the models to actual options market prices to recover the implied volatility surface. The well-known volatility smile — where out-of-the-money puts are priced at higher implied vols than at-the-money options — directly reflects the fat tails and negative skew documented here, and represents the market's endogenous correction to Black-Scholes' distributional assumptions.

References

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